PRICES VERSUS QUANTITIES FOR GREENING FLEXIBLE COMPUTING: ALLOCATION EQUIVALENCE, REVENUE DIVERGENCE, AND THE LONG-RUN RANKING

TEAM A1 — ROLE 1

Abstract. Two instruments can stop a flexible computing load from burning fossil fuel: a carbon tax τ on fossil output, and a dispatch cap that permits the load to run only on realized renewable surplus. At their respective greening thresholds the two are *allocation-equivalent*: identical computing, identical zero carbon from the load, and — as the Guide observes — an identical marginal-entrant condition. We show the equivalence is fragile in exactly the dimension the policy debate cares about. The tax delivers strictly positive revenue to renewables in the hours the load induces; the cap delivers none. Because that revenue enters the renewable first-order condition additively, the two instruments induce *different* equilibrium renewable capacity, and the resulting consumer carbon is strictly ordered. Short-run equivalence therefore does not survive endogenous investment, and the ranking favours the price instrument for a reason Weitzman's calculus does not see: not relative slope uncertainty, but who is paid. We also close the boundary problem: at the greening tax exactly, producers are indifferent and the equilibrium quantity is a correspondence, not a point; the $\tau \downarrow \bar{\tau}$ limit selects the fossil-off equilibrium, which is the selection the comparison requires.

1. Setting

We work in the deregulated (spot) environment, where the relevant computing margin is $R-c-\tau$ rather than $R-p-\tau$. A tax τ per fossil MWh raises the effective fuel cost to $c+\tau$; every formula of the baseline applies under that substitution. In particular the no-load fossil FOC $(\beta+p-c-\tau)[1-F(K_F^\circ)] = C_F$ gives the benchmark reliability

$$\varphi^\circ(\tau) = 1 - \frac{C_F}{\beta + p - c - \tau},$$

and we maintain $\beta + p > R + C_F$ so this stays positive up to

$$\bar{\tau} := R - c,$$

the tax at which a miner-valued MWh no longer covers taxed fuel. Write $q^\circ(\tau) = F(0; K_R^\circ(\tau))$ for the no-load surplus probability. The *cap* (renewable-only contract) is $d(x) \le (-x)_+$: the load may run only on realized surplus.

2. Allocation equivalence at the greening thresholds

Lemma 1 (Viability splits into a taxed and a tax-exempt part)**.** *Under the tax the first entrant is dispatched whenever spare capacity exists. In surplus states (probability q°) it buys at $P = 0$ and keeps R; in fossil-topped states (probability $\varphi^\circ(\tau) - q^\circ$) it buys at $c+\tau$ and keeps $R-c-\tau$. Hence*

$$\bar{C_M}(\tau) = \underbrace{(R-c-\tau)\big[\varphi^\circ(\tau) - q^\circ\big]}_{\textit{fossil-hour part}} + \underbrace{R\,q^\circ}_{\textit{surplus part}} .$$

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Theorem 2 (Allocation equivalence)**.** *At $\tau = \bar{\tau}$ the fossil-hour part of $\bar{C_M}$ vanishes and $\bar{C_M}(\bar{\tau}) = R\,q^\circ$, which is exactly the cap's entry threshold: under either instrument the first entrant earns R in surplus hours only. Consequently, at $(\tau = \bar{\tau})$ versus (cap), holding capacities fixed:* (i) *the computing dispatch schedule coincides ($d = (-x)_+ \wedge K_M$),* (ii) *carbon attributable to the load is zero under both, and* (iii) *the marginal-entrant condition is the same, so the same computing capacity enters. The two instruments admit the same computing at the margin.*

Proof. (i) At $\tau = \bar{\tau}$ fossil-served computing earns $R - c - \bar{\tau} = 0$ and is weakly dominated; the selection of Theorem **??** removes it, leaving surplus-only dispatch, which is the cap's schedule. (ii) Immediate from (i): the load draws no fossil. (iii) Lemma **??** at $\tau = \bar{\tau}$ gives $\bar{C_M}(\bar{\tau}) = Rq^\circ$; under the cap the entrant runs only in surplus hours and earns R there, so its threshold is Rq° as well. □

3. The boundary: indifference, and the selection that resolves it

At $\tau = \bar{\tau}$ producers are exactly indifferent about serving computing from fossil, so the market-clearing *quantity* is not pinned by optimization alone. This is the gap flagged in review, and it must be closed before Theorem **??** can be used.

Theorem 3 (Price is unique, quantity is a correspondence, and the limit selects)**.** *Fix capacities and a state $x \in (-K_M, 0]$ with fossil headroom.*

(a) *For $\tau < \bar{\tau}$: fossil tops up at marginal cost $c + \tau < R$, so $P = c + \tau$ and fossil-served computing is strictly profitable; the equilibrium is unique.*
(b) *At $\tau = \bar{\tau}$: $c + \tau = R$, so the clearing price is* unique*, $P = R$; but every fossil-to-computing quantity in $[0,\, K_M - (-x)_+]$ is consistent with producer indifference and market clearing. The equilibrium* quantity *is a correspondence.*
(c) *For $\tau > \bar{\tau}$: fossil-served computing is strictly unprofitable, so the unique equilibrium has fossil-to-computing quantity* 0*.*

Consequently the correspondence at $\bar{\tau}$ is the upper limit of (a) and the lower limit of (c), and the refinement $\tau \downarrow \bar{\tau}$ selects the fossil-off point. We adopt that selection throughout; equivalently, one may state all results for $\tau > \bar{\tau}$ and pass to the limit, which is the conservative reading.

Proof. (a) and (c) are immediate from the sign of $R - c - \tau$. For (b), at $c + \tau = R$ a fossil generator earns exactly zero on a computing MWh and a miner earns exactly zero, so any split clears; the price cannot exceed R (demand vanishes above it) nor fall below $c + \tau$ (supply vanishes below it), and these coincide, pinning $P = R$. The limit statement follows because the fossil-off quantity is the unique selection on $(\bar{\tau}, \infty)$ and the correspondence is upper hemicontinuous at $\bar{\tau}$. □

Remark 4. The same argument disposes of the companion objection that "the price never leaves 0" under the cap. Under the cap, miner demand in surplus states equals the zero-marginal-cost surplus exactly, so any $P \in [0, R]$ clears; the competitive selection is $P = 0$ because supply is perfectly elastic at zero up to the surplus. That is a selection, not an implication, and we flag it as such.

4. Revenue divergence, and why the equivalence dies in the long run

Proposition 5 (Revenue divergence)**.** *Per unit of renewable capacity per hour, the two instruments deliver*

$$\textit{tax:}\quad \mathbb{E}\big[\tilde{S}\,\min\{c+\tau, R\}\,\mathbf{1}_{\{-K_M < \tilde{x} \le 0\}}\big] > 0, \qquad \textit{cap:}\quad 0.$$

The tax's pass-through to renewable revenue is nondecreasing in τ and saturates at $\bar{\tau}$; the cap leaves surplus-hour prices at zero by construction, because miner demand there exactly equals the zero-marginal-cost surplus.

This is the crux. The two instruments buy the same allocation at different prices, and the difference is paid to renewable capacity.

Theorem 6 (Long-run non-equivalence and a strict ranking)**.** *Let K_R^{tax} and K_R^{cap} denote equilibrium renewable capacity under the greening tax and under the cap. The renewable first-order conditions differ by exactly the strictly positive term of Proposition* **??***; hence, whenever the renewable marginal value is strictly decreasing (the standing single-peakedness),*

$$\boxed{K_R^{\text{tax}} > K_R^{\text{cap}}, \qquad \textit{and therefore} \qquad \mathbb{E}[\textit{fossil}^{\text{tax}}] < \mathbb{E}[\textit{fossil}^{\text{cap}}].}$$

Allocation equivalence at fixed capacities does not *survive endogenous investment: the instruments are strictly ranked, and the price instrument dominates.*

Proof. Both instruments induce the same computing dispatch (Theorem **??**), so the consumer-side terms of the renewable FOC coincide. The tax adds the surplus-hour revenue of Proposition **??**, which is strictly positive whenever $K_M > 0$ and $\Pr(-K_M < \tilde{x} \leq 0) > 0$. Evaluating the tax economy's renewable marginal value at K_R^{cap} therefore exceeds C_R; by single-peakedness the tax optimum lies strictly to the right. The carbon conclusion follows from $e(K_R) = \int_0^{K_F(K_R)}(1-F)$ being strictly decreasing in K_R. □

Corollary 7 (What the ranking is *not*)**.** *The ranking is not an efficiency comparison in Weitzman's sense: it does not turn on the relative slopes of marginal benefit and marginal cost under uncertainty. Both instruments implement the same allocation. The wedge is* distributional *— who receives the scarcity rent in surplus hours — and it becomes an efficiency difference only because that rent is also the marginal return to renewable investment. A quantity instrument that rebated the same rent to renewables would replicate the tax.*

Remark 8 (Policy reading). New York's moratorium on fossil-fired proof-of-work mining, which exempts renewable-powered operations, is a cap. Theorem **??** says that such a rule buys the same short-run decarbonization as the equivalent carbon price and forgoes the renewable build-out the price would have financed. The cap is the feasible near-term instrument at a known cost, and the cost is quantified here rather than asserted.